

EXP 3

IMPLEMENTING MINIMAX ALGORITHM

```
import math

def minimax(depth, nodeIndex, isMax, scores, h):
    if depth == h:
        return scores[nodeIndex]

    if isMax:
        return max(
            minimax(depth + 1, nodeIndex * 2, False, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, False, scores, h)
        )
    else:
        return min(
            minimax(depth + 1, nodeIndex * 2, True, scores, h),
            minimax(depth + 1, nodeIndex * 2 + 1, True, scores, h)
        )

scores = [3, 5, 2, 9, 12, 5, 23, 23]

h = int(math.log2(len(scores)))

print("The optimal value is:", minimax(0, 0, True, scores, h))
```

OUTPUT:

The optimal value is 12